VOLT-FORGE

Build the circuit. See the consequence.

A connected workspace to explain, challenge and repeat battery design decisions

01 Define the job	02 Change the design	03 Challenge the answer	04 Keep the evidence
Pi 5 gateway · 25 W supply scenario · 10 min	Cells, wiring, cooling and initial state	Mismatch, losses and operating boundaries	Saved inputs, model, results and export

PROBLEM UNDERSTANDING · DESIRABILITY

Can a smaller backup still do the job?

Fewer cells cost less; each works harder. Wiring and cooling also change the answer. Students can see the interaction; engineers can compare candidates before a bench test.

TECHNICAL FEASIBILITY I Required sensitivity cases

Same-end design	Cases passed	Worst cell current	Worst cell temperature	Cell cost
4 cells	51/51	3.34 A	26.1 °C	₹4,824
3 cells	42/42	3.86 A	27.7 °C	₹3,618
2 cells	33/33	5.49 A	31.8 °C	₹2,412

Same job and settings. Case counts vary with cell positions. Cost covers cells only; temperatures are simulated.

Two cells passed 33/33 required cases; one was refused. Cell cost and mass halve, while electrical margin narrows.

ENGINEERING QUALITY

Keep the conclusion conditional

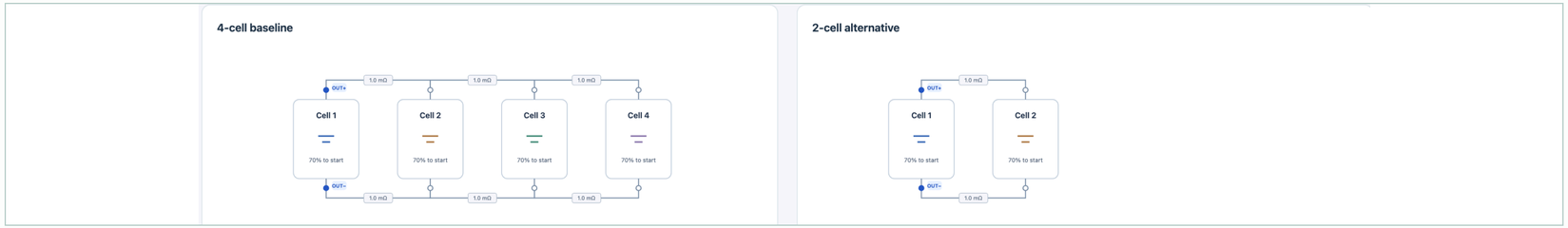
Numerical checks pass. All 12 comparisons with published measurements missed voltage and temperature bands. Predictive accuracy and board compatibility remain open. Two cells have no spare.

PROTOTYPE & EVIDENCE · INNOVATION

Test it out: volt-forge-app.vercel.app



Build it. An editable circuit, connected to actual cell properties and test assumptions.



Compare it. Pressure tests, monitoring and virtual cutoffs challenge the design; physical protection is unverified.

4,006 cases reproduced
No stored result changed in the final rerun.

54 published experiments
194 files checksum-verified; 2 module records compared.

15/15 numerical checks
Electrical and heat calculations against PyBaMM.

23 °C · 70% charge · assumed 90% converter and 3.3 V floor · fixed 8.42 A proxy · one parallel group; ageing/packaging later

BUSINESS VIABILITY

Students free. Labs ₹50,000/year; teams ₹5 lakh/year. Prices untested. AISHE 2021–22: 41.3 lakh E&T students; 15.6 lakh in three target disciplines; 130 labs at 5% assumed reach.

Calculations & sources
volt-forge-app.vercel.app/appendix